COCO-Ghost: Auditing Whether VLM Object Claims Survive Object Removal

Anusha Kanagala, M.S.; Dhiral Panjwani, M.S.; Stephanie M. Aguilera, B.S.; Abhiraj Pudhota, M.S.; Kelly Powell, M.B.A.; Danish Murad, M.S.; Rubin Pillay, M.D.; Leon Jololian, Ph.D.; and Sandeep Bodduluri, M.S., Ph.D.

Abstract

Vision-language models can report objects that are contextually plausible but visually absent. COCO-Ghost evaluates this behavior with object-level counterfactuals: an object must be recognized in the original image and in its crop, then the same object is occluded and queried again. If the model continues to answer YES after clean target removal, the claim is counted as a ghost. On 200 audited COCO validation instances, three local VLMs retain removed-object claims in 33.7-51.6% of originally recognized cases. GHOST-Guard is a training-free, black-box abstention rule that accepts object claims only when they are supported before removal and disappear after removal.

Core idea

The diagnostic asks a deliberately narrow question: if a model says an object is visible, does that claim depend on local visual evidence or only on scene context? This is useful because context priors can be semantically reasonable while still being visually unsupported.

- Original query: ask whether the target object is present in the full COCO image.
- Crop gate: ask the same question on a crop around the target object.
- Masked query: replace the target region with padded bounding-box solid local-mean fill and ask again.
- Ghost criterion: original YES and masked YES after target removal.
- GHOST-Guard accept criterion: original YES, crop YES, and masked NO.

Scaled clean-removal results

The headline run uses 200 COCO validation instances across 20 object categories and three local Ollama VLMs. All 200 counterfactuals are labeled clean-removal. The solid bbox intervention is less naturalistic than blur, but it avoids object-shaped blur that could leak the target silhouette.

Model	Original YES	Crop YES	Masked YES	Ghost rate	Guard accept
qwen2.5vl:7b	96.0%	94.5%	49.5%	51.6%	44.5%
qwen2.5vl:3b	88.5%	80.0%	43.0%	47.5%	30.0%
llava:7b	43.0%	60.5%	14.5%	33.7%	24.5%

Ghost rate is computed among original-YES cases. Guard accept is the fraction of all evaluated cases accepted after the counterfactual test. Detected ghost claims accepted by GHOST-Guard were 0.0% in all three model runs.

Qualitative evidence and artifact audit

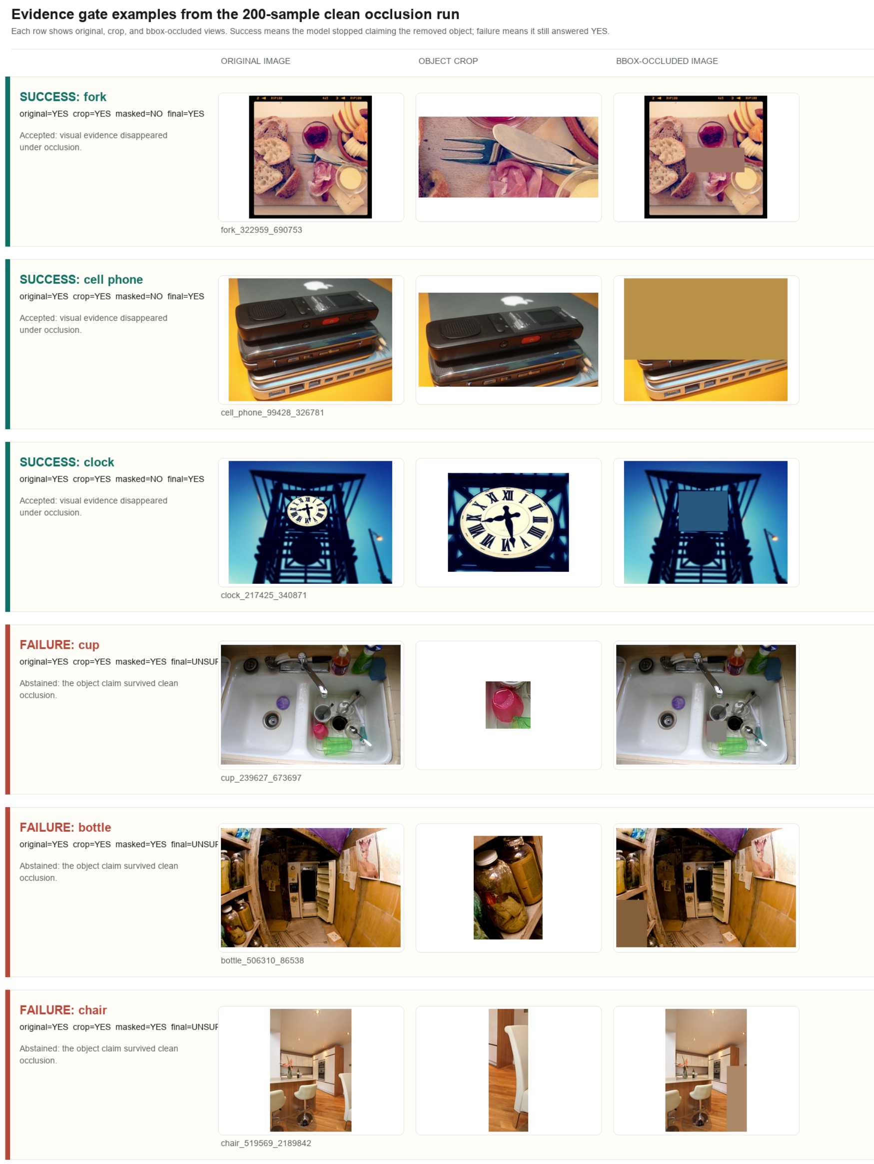

Figure 1. Representative success and failure cases. A failure means the masked view still receives YES.

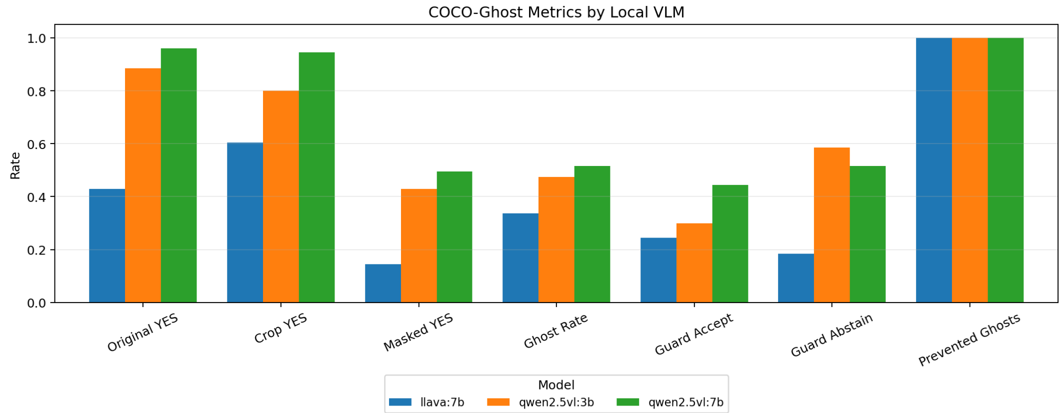

Figure 2. Aggregate metrics across three local VLMs on the same 200 clean counterfactuals.

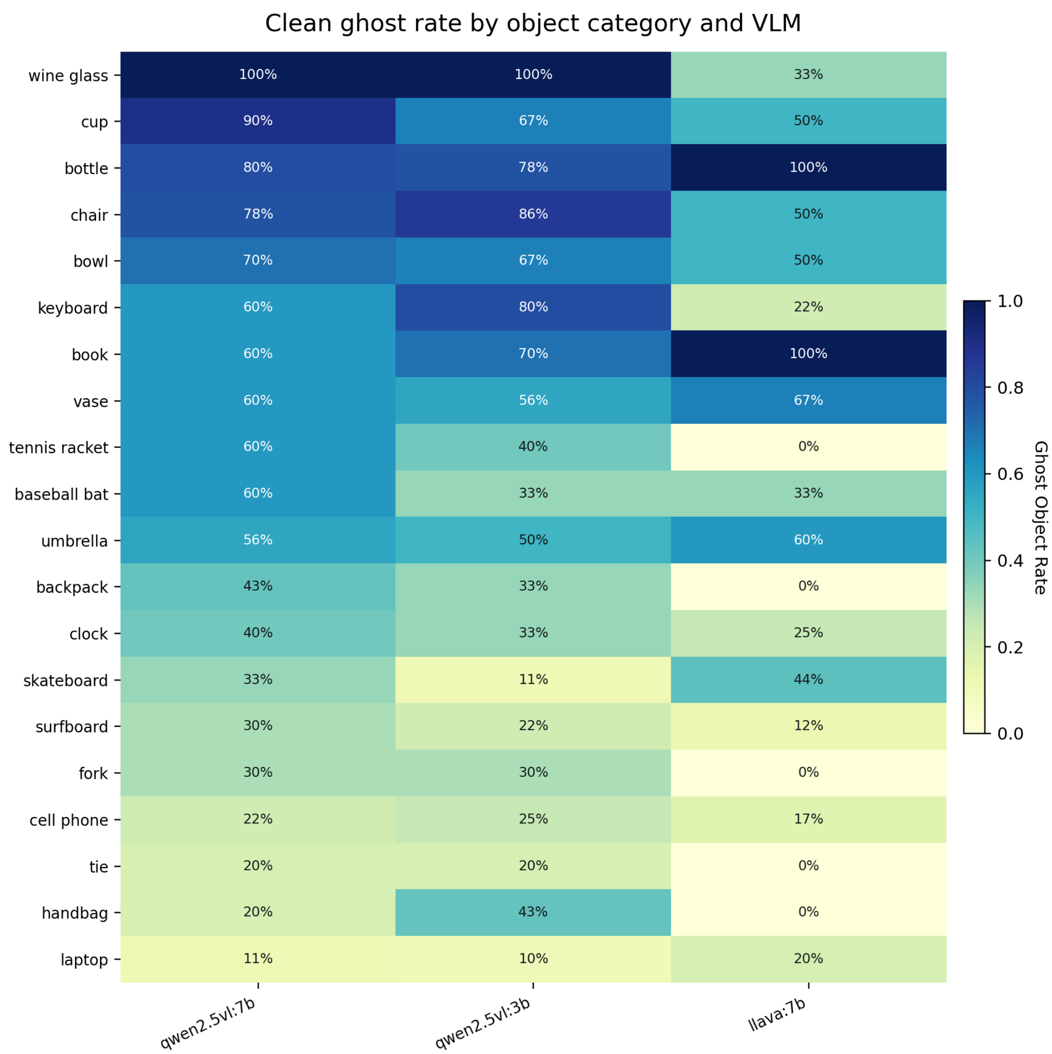

Figure 3. Category-by-model ghost rates reveal object classes with strong context persistence.

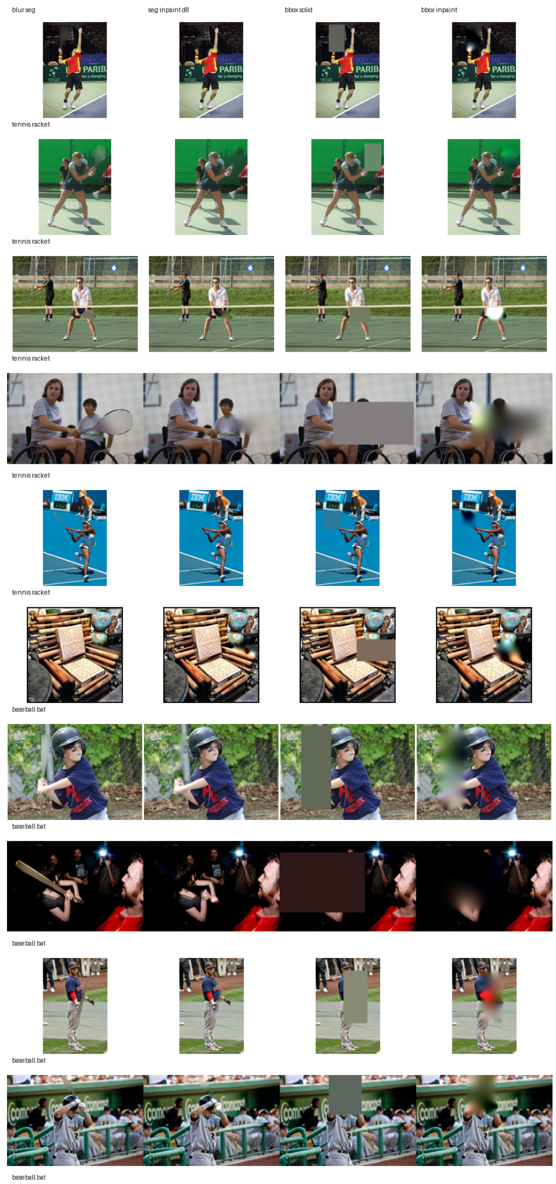

Figure 4. Removal-method audit. Bbox-solid occlusion reduces object-shaped residual evidence.

Interpretation and limitations

GHOST-Guard is not a mechanistic explanation and does not repair the underlying model. It is a black-box reliability wrapper for object-presence claims. In deployed settings, abstentions can be routed to a detector, a stronger model, another view, or human review.

COCO-Ghost is a counterfactual stress test for one interpretable claim type: object presence. It does not prove that every masked-image YES is a human-level hallucination, and it does not replace broader VLM robustness evaluation.

Citation

```
@misc{cocoghost2026,
  title = {COCO-Ghost: Auditing Whether VLM Object Claims Survive Object Removal},
  author = {Kanagala, Anusha and Panjwani, Dhiral and Aguilera, Stephanie M. and Pudhota, Abhiraj and
Powell, Kelly and Murad, Danish and Pillay, Rubin and Jololian, Leon and Bodduluri, Sandeep},
  year = {2026},
  url = {https://github.com/DIGlabUAB/coco-ghost-guard}}
```